

Environmental monitoring

IOT CLASS PROJECT 2024-2025,
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PROF. FRANCESCO PUPO

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CALABRIA

DIPARTIMENTO DI INGEGNERIA
INFORMATICA, MODELLISTICA,
ELETTRONICA E SISTEMISTICA

DIMES

Content

- **Goals**
- **Architecture**
- **Context-aware**
- **Challenges**
- **Features**
- **Used technologies**
- **Future development**

Goals

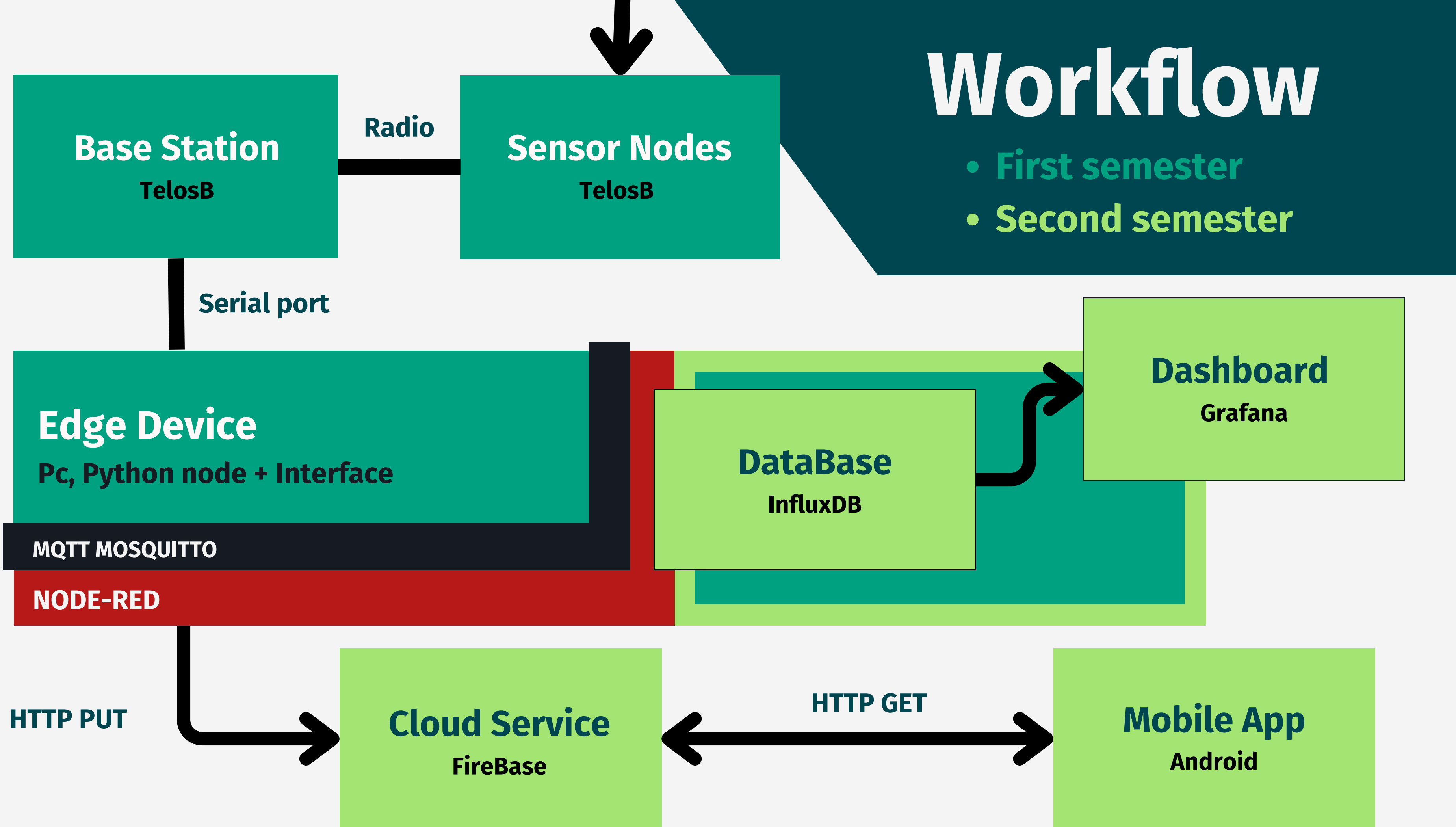


Main Goal:

**Museum's environment
Monitoring Dashboard**

Side Goal:

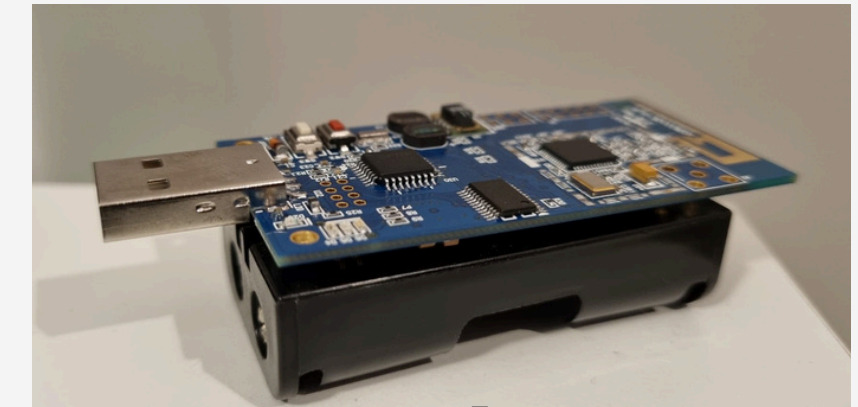
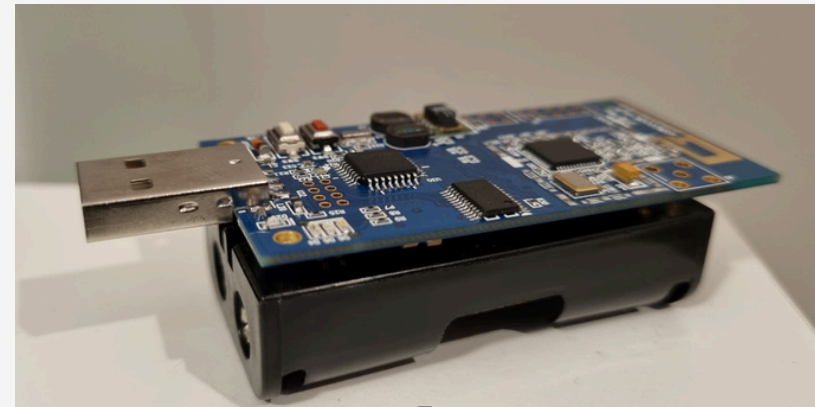
**Museum's environment
Realtime updating**





Sensor Nodes

TelosB



radio

Base Station

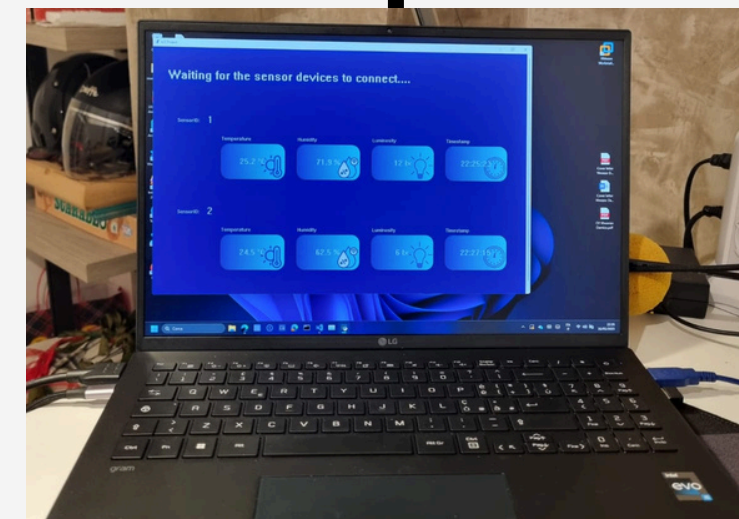
TelosB



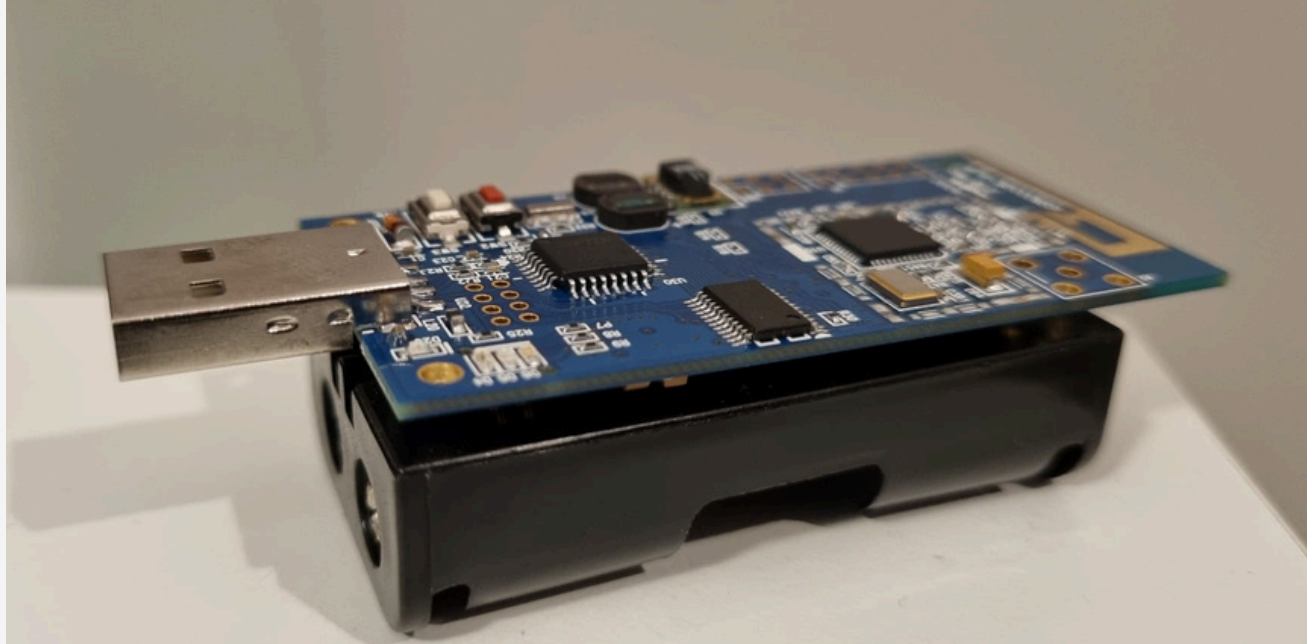
Serial Port

Edge Device

PC, Python Interface



Sensor Nodes



2 TelosB



MAKEFILE

SensingAppC.nc

SensingC.nc

SensorMsg.h

Base Station



1 TelosB



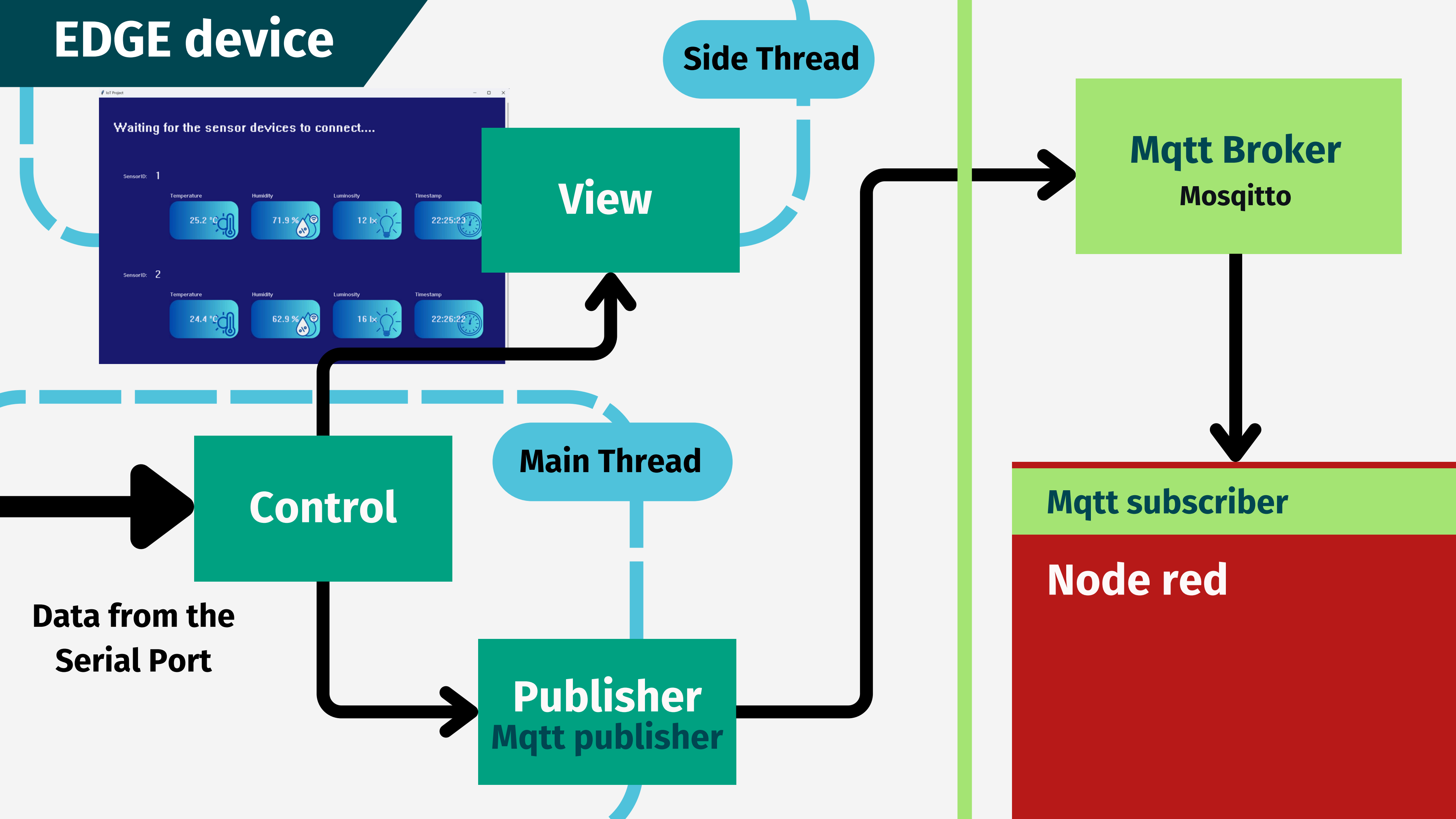
MAKEFILE

**SimpleReceiver
AppC.nc**

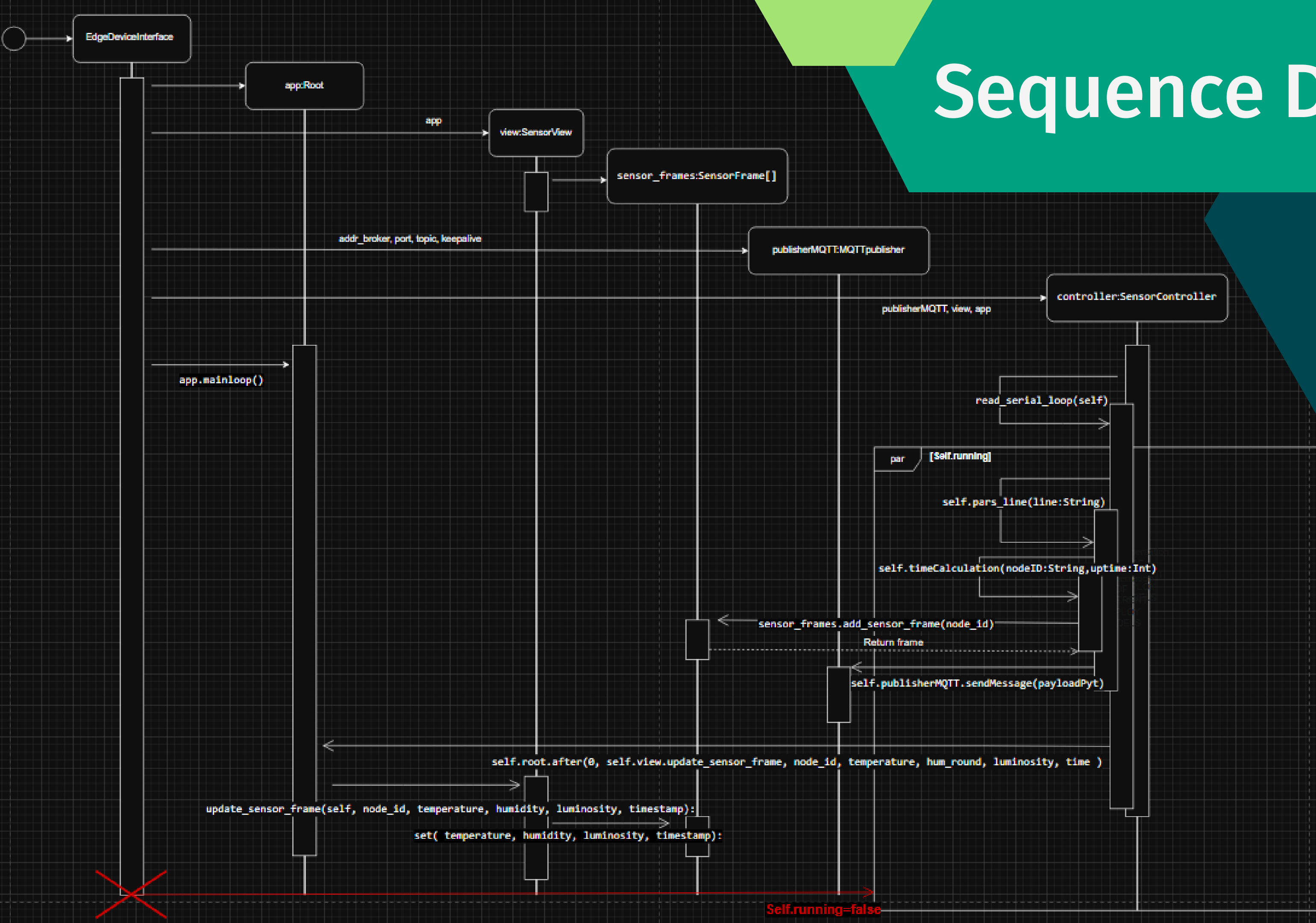
SimpleReceiverC.nc

SensorMsg.h

EDGE device



Sequence Diagram



EDGE device

Mqtt QoS1

Node red

Mqtt subscriber

Adaptation of the
Node-message for
Firebase

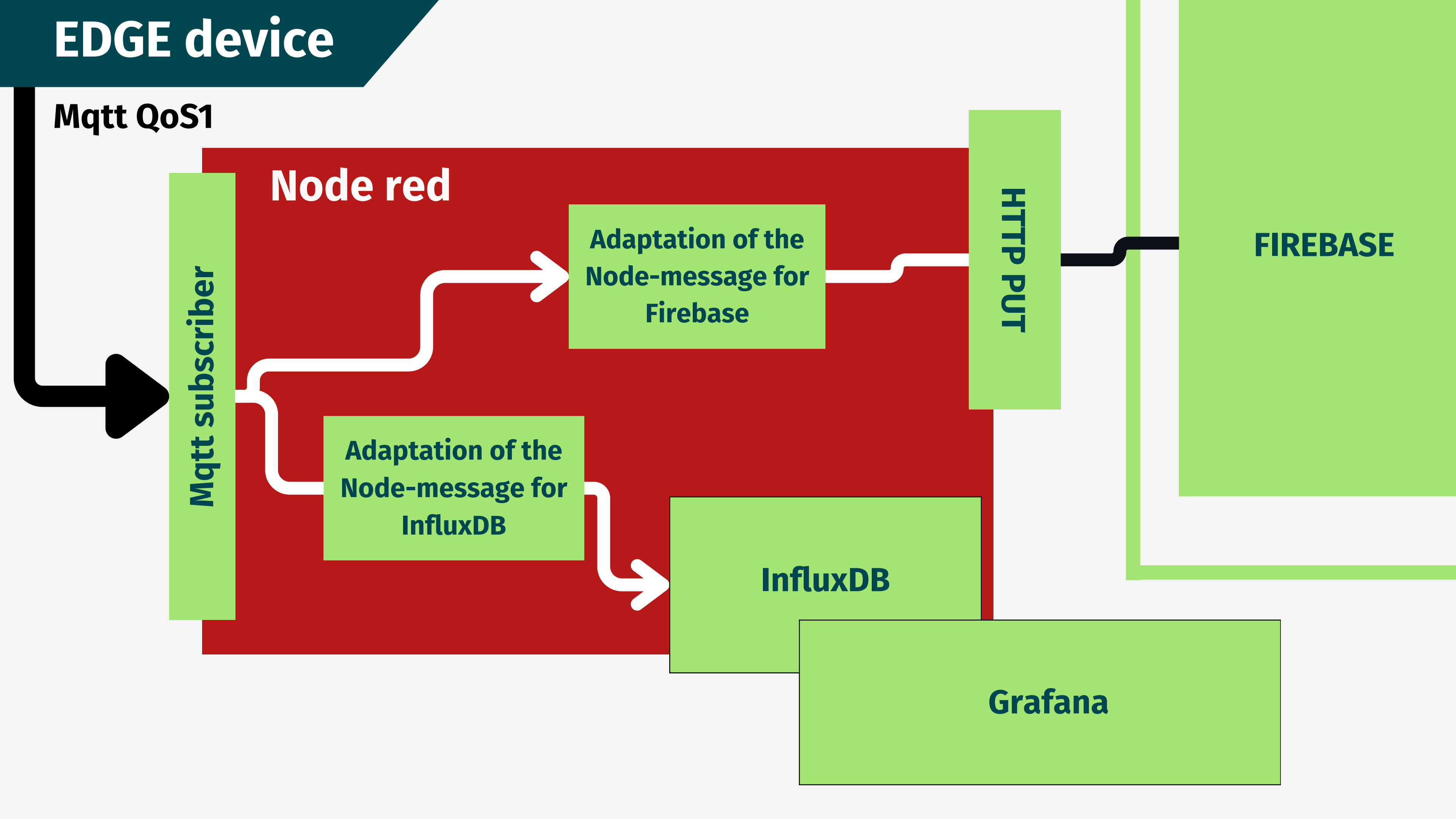
HTTP PUT

FIREBASE

Adaptation of the
Node-message for
InfluxDB

InfluxDB

Grafana



EDGE device

Mqtt QoS1

Node red

```
1 let message = msg.payload;
2 msg.payload = {
3   temperature: message.temperature,
4   humidity: message.humidity,
5   luminosity: message.luminosity,
6 };
7
8 msg.url = "https://iot-environmental-14e42-default-rtdb.europ
9   message.idSensor + "/" + message.timestamp+".json";
10 return msg;
```

Adaptation of the
Node-message for
Firebase

HTTP PUT

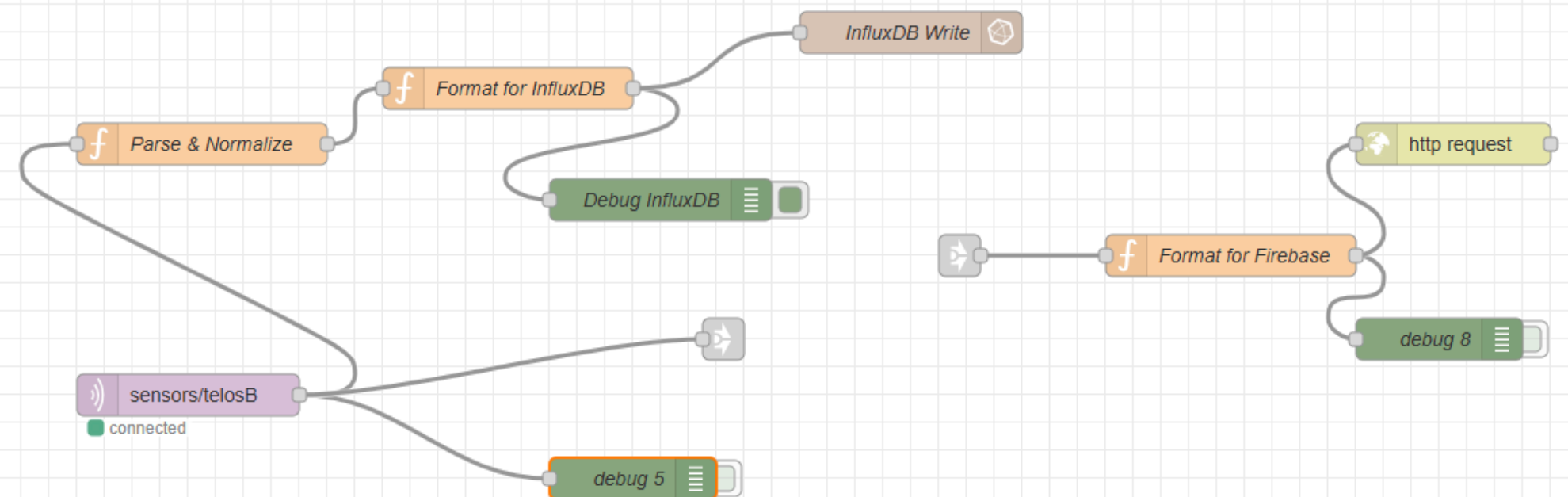
FIREBASE

Mqtt subscriber

Adaptation of the
Node-message for
InfluxDB

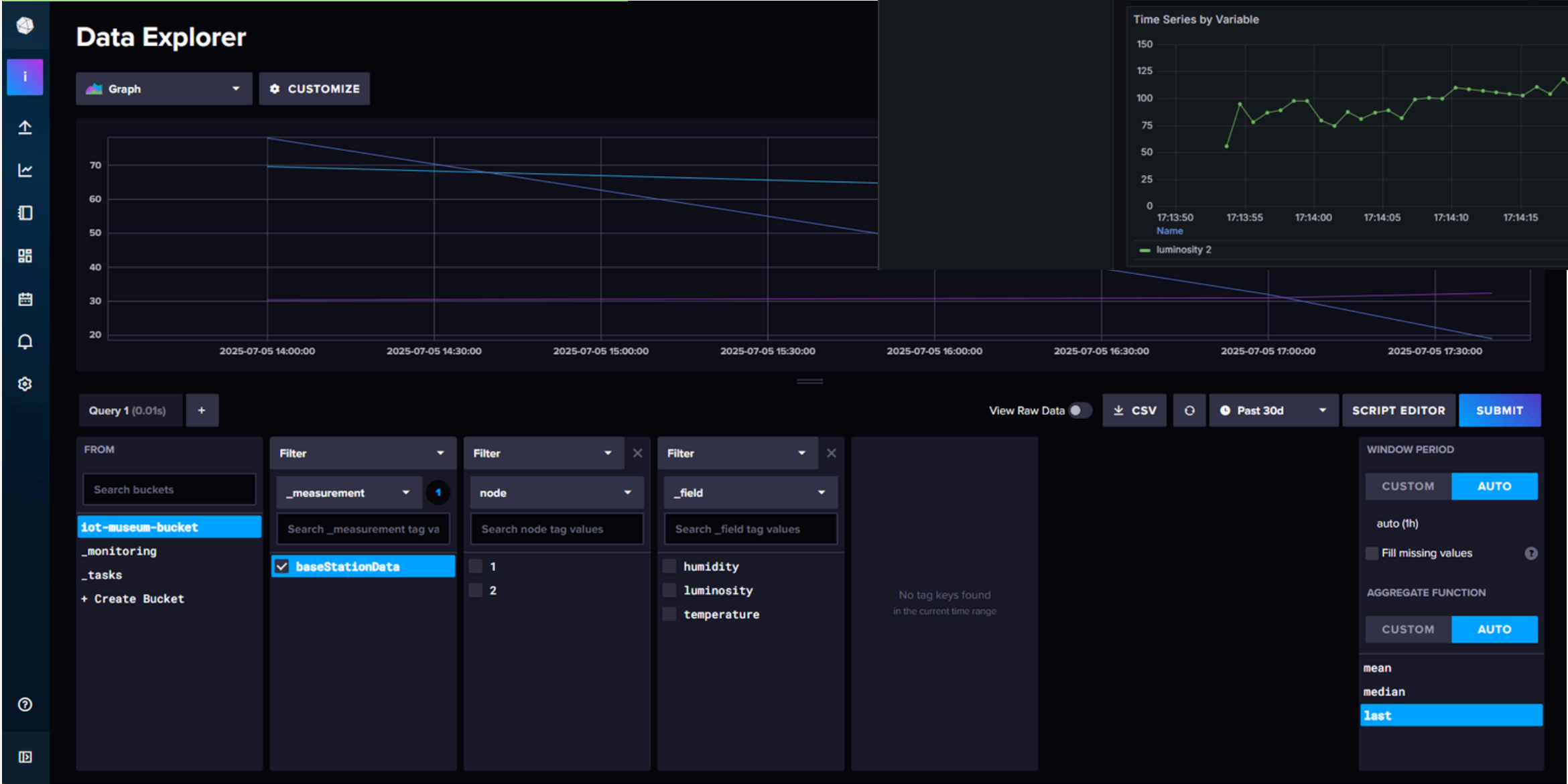
Flow

```
1 msg.payload = [
2   {
3     temperature: msg.temperature,
4     humidity: msg.humidity,
5     luminosity: msg.luminosity
6   },
7   {
8     node: msg.sensorId.toString()
9   }
10 ];
11
12 msg.timestamp=msg.timestamp;
13 return msg;
```



EDGE device

http://localhost:8086



http://localhost:3000

FIREBASE



```

baseStation
├── 1
│   ├── 1
│   │   ├── 2025-06-30_10:09:22
│   │   │   ├── humidity: 55.8
│   │   │   ├── luminosity: 628
│   │   │   └── temperature: 28.9
│   │   ├── 2025-06-30_10:09:23
│   │   └── 2025-06-30_10:09:24
│   └── 2
│       └── 1
│           ├── 2025-06-30_14:38:09
│           ├── 2025-06-30_14:38:10
│           └── 2025-06-30_14:38:18
└── 2
    └── 1
        ├── 2025-06-30_14:38:09
        ├── 2025-06-30_14:38:10
        └── 2025-06-30_14:38:18
  
```

Data

Data Representation

SensorReading
 NodeData
 BaseStationData

MAIN ACTIVITY

View

Screen

Model

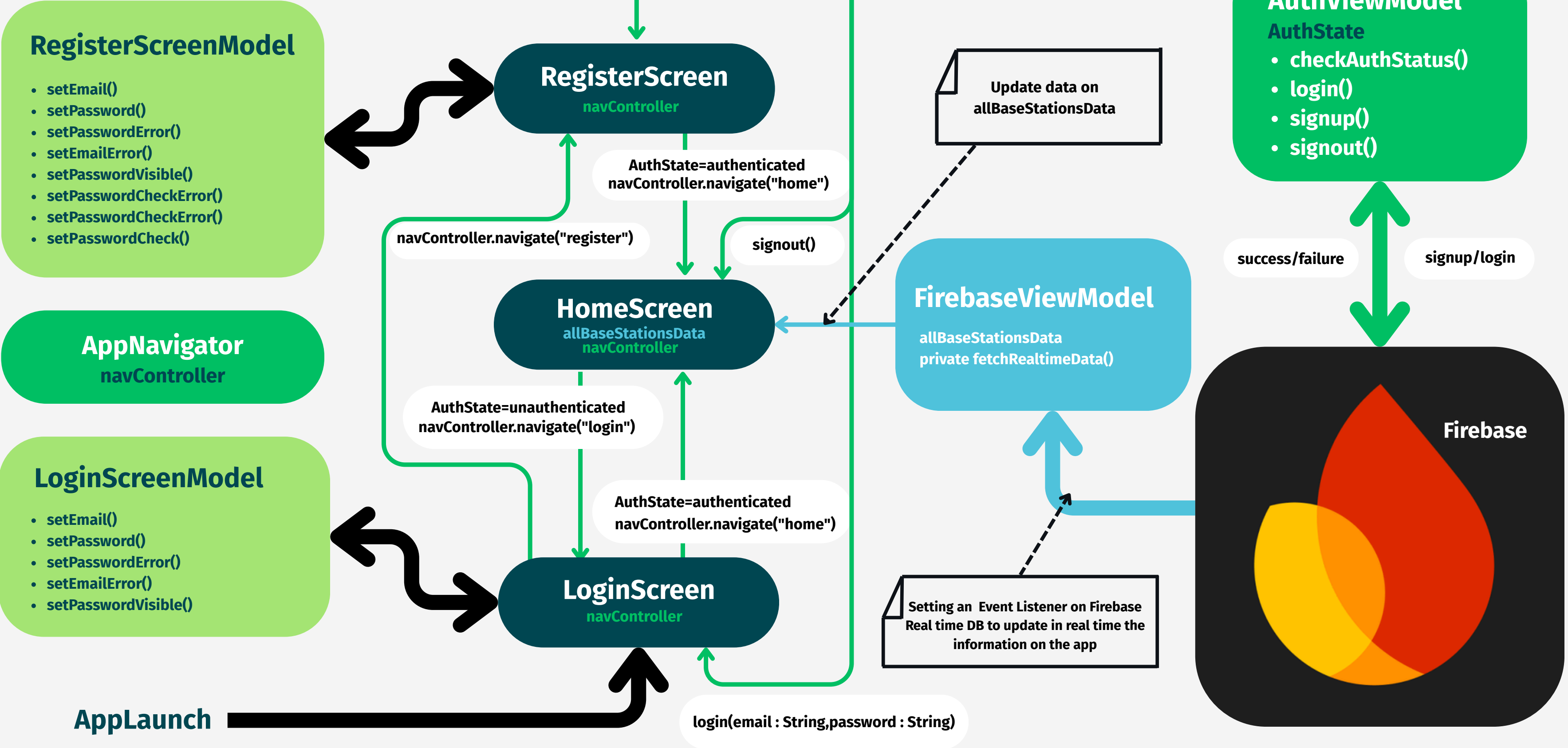
HomeScreen
 LoginScreen
 RegisterScreen

AuthViewModel
 FirebaseViewModel
 LogicViewModel
 RegisterViewModel

Nav

AppFragmentNavigator

Firebase + Android



Android UI

19:12 9



Create an account

Email

Password

Password Confirmation

Sign up

Already have an account? [Login!](#)

19:12 9



Email

Password

Login

Not a member? [Sing up now!](#)

19:18 9

Node List

Base Station: 1

Node ID: 1

Timestamp: 2025-06-30 11:39:36
Temperature: 30.5°C
Humidity: 53.2%
Luminosity: 179lx

Node ID: 2

Base Station: 2

Node ID: 1

Node ID: 2

Sign out

19:18 9

Node List

Base Station: 1

Node ID: 1

Timestamp: 2025-06-30 11:39:36
Temperature: 30.5°C

Sign out

19:18 9



Create an account

Email

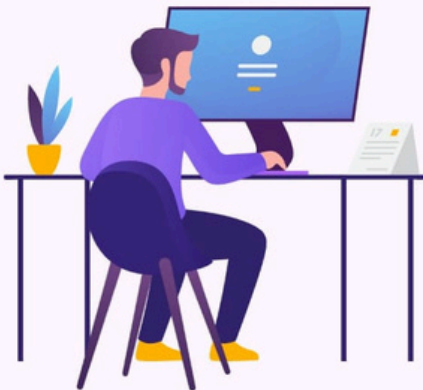
Password

Password Confirmation

Sign up

Already have an account? [Login!](#)

19:18 9



Login

Email

Password

Login

Not a member? [Sing up now!](#)

Context-aware

*Scalable



2* ~ TelosB sensors to monitor temperature, humidity and luminosity

1* ~ TelosB base station

1* ~ PC that act EdgeDevice

1* ~ Android mobile Device



Challenges

- **Measurement Frequency**
- **Radio communication fails between TelosB**
- **Sensors Placement**
- **Uncertainty**



Challenges

- **MQTT implementation with QoS=1 to balance RealTime constraints & Reliability**
- **Flow management**



Challenges

- **Integration of Node-red and InfluxDB**
- **Integration of InfluxDB and Grafana**



Challenges

- **Design following the JetPackCompose Paradigm**
- **Real time synchronization with the sensors datas on Firebase**

Features

Dashboard that provides real-time visualization of the sensors datas

Real time synchronization with Android mobile devices

Cloud integration to store and share data

Easy scalable system

What did we use?

TinyOS (.nc)	Python
Firebase	MQTT (Nodered+ Mosquitto)
Grafana	Android

Future development

Implementation of actuation to change temp, hum, lum	Reinforced learning model for auto-regulation of the actuators
Implementation of area data inference model	Advise deployment AI System for better positioning of the sensors

Git-Hub



Network graph

Timeline of the most recent commits to this repository and its network ordered by most recently pushed to.



https://github.com/VincenzoDamico/IOT_TinyOs

Thank you

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